

UNITED INSTITUTE OF TECHNOLOGY
DATA STRUCTURE(BCS-301)
ASSIGNMENT 4
BASED ON PREVIOUS YEAR SEMESTER PAPERS' QUESTIONS
UNIT 4

SECTION A

1. Attempt all questions in brief:

- a) Discuss the concept of "successor" and "predecessor" in Binary Search Tree.
- b) Explain height balanced tree. List general cases to maintain the height.
- c) Number of nodes in completely 100000. Find its depth.

the number of nodes in a complete binary tree of depth (d) is $(2^{d+1} - 1)$.

For (100,000) nodes:

- $(2^{d+1} - 1 = 100,000)$
- $(2^{d+1} = 100,001)$

To find the depth, we need to solve for (d) using logarithms:

$$(\log_2(2^{d+1}) = \log_2(100,001)) \quad (d+1 = \log_2(100,001)) \quad (d = \log_2(100,001) - 1)$$

Let's calculate that:

$$[d = \log_2(100,001) - 1 \approx 16.609]$$

Since depth should be an integer value in this context (as it represents levels), the depth of a complete binary tree with (100,000) nodes is (16).

- d) Define extended binary tree, full binary tree, strictly binary tree and complete binary tree.
- e) Explain threaded binary tree.
- f) Explain the minimum number of interchanges needed to convert the array. 90, 20, 41, 18, 13, 11, 3, 6, 8, 12, 7, 71, 99, into a maximum heap.
- g) What is the importance of threaded binary tree?
- h) Write short notes on mini heap.
- i) Draw the binary search tree that results from inserting the following numbers in sequence starting with 11: 11, 47, 81, 9, 61, 10, 12
- j) Write advantages of AVL tree over Binary Search Tree (BST)
- k) In a complete binary tree if the number of nodes is 1000000. What will be the height of complete binary tree.
- l) Illustrate the significance of Threaded Binary Tree.

SECTION B

2. Attempt all questions in brief:

- a) Define tree, binary tree, complete binary tree and full binary tree. Write algorithms or function to obtain traversals of a binary tree in preorder, postorder and inorder.
- b) Construct a B-tree on following sequence of inputs. 10, 20, 30, 40, 50, 60, 70, 80, 90. Assume that the order of the B-tree is 3.
- c) Explain Huffman algorithm. construct Huffman tree for MAHARASHTRA, with its optimal code.
- d) What is height balanced tree? Why height by balancing of tree is required? Create an AVL tree for the following elements: a,z,b,y,c,x,d,w,e,v,f.
- e) What is the difference between binary search tree and heap? For a given sequence of numbers, construct a heap and a BST. 34,23,67,45,12,54,87,43,98,75,84,93,31.
- f) The order of nodes of a binary tree in inorder and postorder traversal are as follows:
In order : B, I, D, A, C, G, E, H, F.
Post order: I, D, B, G, C, H, F, E, A.
 - (i) Draw the corresponding binary tree.
 - (ii) Write the pre order traversal of the same tree.

Construct a binary tree with the following traversals:

Inorder: B C A E G D H F I J

Preorder: A B C D E G F H I J

Construct an AVL Tree by inserting following sequence of elements, starting with an empty tree: 71,41,91,56,60,30,40,80,50,55

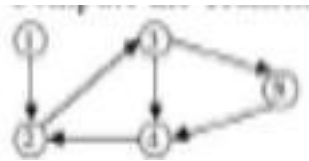
SECTION C

3. (a) Draw a binary tree with following traversals: Inorder: B C A E G D H F I J
Preorder: A B C D E G F H I J
(b) Consider the following AVL Tree and insert 2, 12, 7 and 10 as new node. Show proper rotation to maintain the tree as AVL.
4. (a) Construct a B-tree of order 5 created by inserting the following elements: 3,14, 7,1,8,5, 11, 17, 13,6,23, 12, 20, 26, 4,16, 18, 24, 25, 19. Also delete element 6, 23 and 3 from the constructed tree.
(b) What is a Threaded Binary Tree? Explain the advantages of using a threaded binary tree.
5. (a) Write the algorithm for deletion of an element in Binary search tree
(b) Construct the binary tree for the following:
In-order : Q,B,K,C,F,A,G,P,E,D,H,R
Preorder: G,B,Q,A,C,K,F,P,D,E,R,H
find the post order of the tree.
6. (a) Can you find a unique tree when any two traversals are given? Using the following traversals construct the corresponding binary tree:
In-order: H K D B I L E A F C M J G
preorder A B D H K E I L C F G J M
(b) What is a B-tree? Generate a B-tree of order 4 with the alphabets (Letters) arrive in the sequence as follows A, G, F, B, K, D, H, M, J, E, S, I, R, X, C, L, N T, U, P.
7. (a) Write short notes are following:
I) Extended binary tree II) Complete binary tree III) Threaded binary tree
(b) Insert the following sequence of element into and A, BL tree starting with empty tree. 7141 915660 30 4080 50 55 Also find the minimum array size to represent this
8. (a) Write an iterative function to search a key in Binary Search Tree (BST).
(b) (i) Why does time complexity of search operation in B-Tree is better than Binary Search Tree (BST)?
(ii) Insert the following keys into an initially empty B-tree of order
a, g, f, b, k, d, h, m, j, e, s, i, r, x, c, l, n, t, u, p
(iii) What will be the resultant B-Tree after deleting keys j, t and d in sequence?
9. (a) Discuss left skewed and right skewed binary tree. Construct an AVL tree by inserting the following elements in the order of their occurrence: 60, 2, 14, 22, 13, 111, 92, 86.
(b) What is B-Tree? Write the various properties of B- Tree. Show the results of inserting the keys F, S, Q, K, C, L, H, T, V, W, M, R, N, P, A, B in order into a empty B-Tree of order 5.
10. (a) Demonstrate B-Tree? Construct a B-Tree of order 4 with the alphabets (letters) arrive in the sequence as follows: a g f b k d h m j e s i r x c l n t u p.
(b) Construct a **Binary Search Tree (BST)** using the following sequence of numbers: 50, 30, 70, 20, 40, 60, 80, 35, 45. Perform Inorder traversal of Tree.

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UNIT 5

SECTION – A

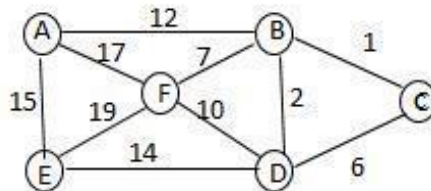
1. (a) List the different types of representation of graphs.
- (b) How the graph can be represented in memory? Explain the suitable example
- (c) What is minimum cost spanning tree? Give its applications.
- (d) Compare adjacency matrix and adjacency list representation of graph.
- (e) Compare the transitive closure of following graph.



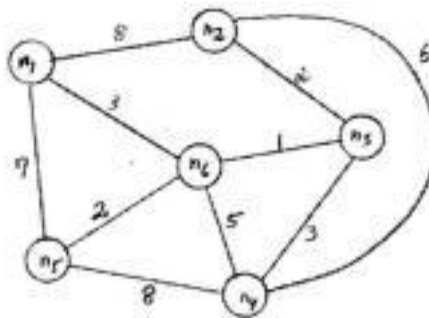
- (f) Write short notes on adjacency multi list representation graph.
- (g) Write different representations of graphs in the memory.
- (h) Write an algorithm for Breadth First Search (BFS) traversal of a graph.
- (i) Explain the concept of NonLinear data structures with the help of example.

SECTION B

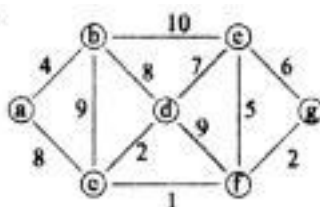
- 2 (a). Define spanning tree. Also construct minimum spanning tree using prim's algorithm for the given graph.



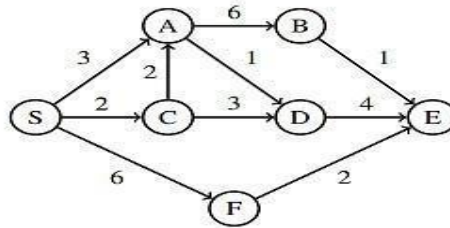
- (b) write the Floyd Warshall algorithm to compute the all pairs shortest path. Apply the algorithm on following graph



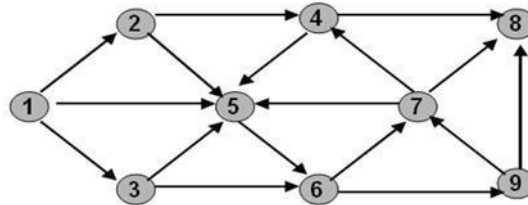
- (c) Find the minimum spanning tree in the following graph using Kruskal's algorithm.



- (d) Write the Dijkstra algorithm for shortest path in a graph and also find the shortest path from 'S' to all remaining vertices of graph in the following graph:



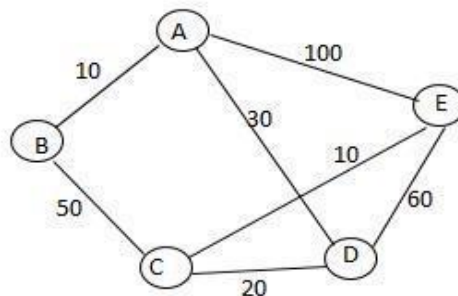
- f) Write an algorithm for breath first search item search(BFS) and explain with the help of suitable example.
g) Differentiate between DFS and BFS. Draw the breadth First Tree for the above graph.



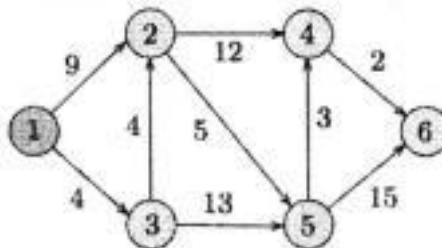
- (g) Explain the Depth-First Search (DFS) algorithm with the help of an example graph. Which data structure is used for DFS and BFS?

SECTION C

- 3 (a) Describe Dijkstra's algorithm for finding shortest path. Describe its working for the graph given below.

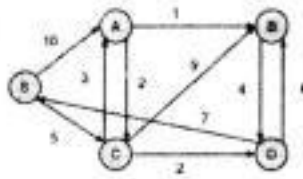


- (b) By considering vertex "1" as source vertex, find the shortest path to all other vertices in the following graph using Dijkstra's algorithm. Show all the steps.



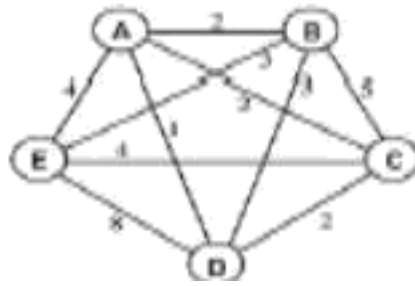
- 4 (a) Explain the detail about the graph traversal technique with suitable examples.
(b) Explain Warshall's algorithm with the help of an example.

5 (a) Describe the Dijkstra algorithm to find the shortest path. Find the shortest path in the following graph with vertex “S” as source vertex.

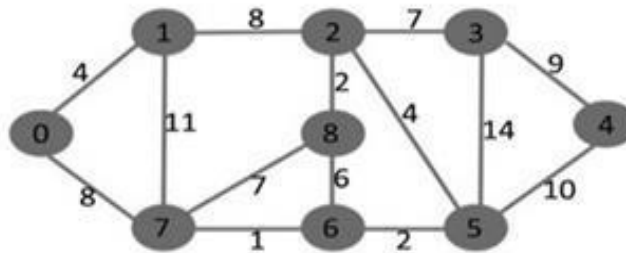


(b) Describe Prim’s algorithm and find the cost of minimum spanning tree using Prim’s algorithm. Explain algorithm with the help of an example.

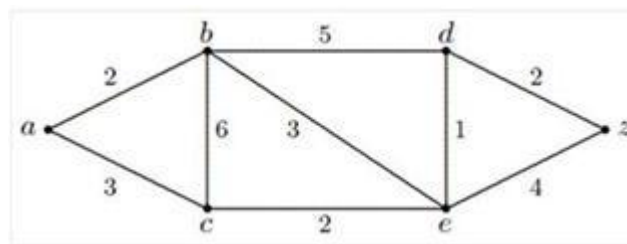
- 6 a. Explain two ways to represent graph in memory and compare their advantages: CO5 K4
- Adjacency matrix
 - Adjacency List
- b. Explain the following graph terminologies with examples:
- Graph
 - Weighted Graph
 - Degree of a Vertex



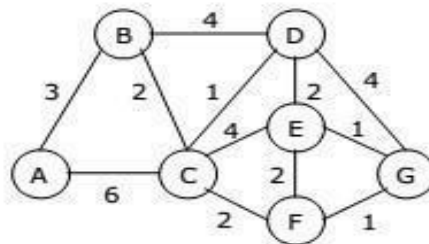
- 5 (a) Apply the Floyd Warshall's algorithm in above mentioned graph (i.e. in Q. no. 6a)
 (b) Use Dijkstra's algorithm to find the shortest paths from source to all other vertices in the following graph.



- 7 (a) Apply Prim's algorithm to find a minimum spanning tree in the following weighted graph as shown below.



- (b) What is spanning tree? Write down the Prim's algorithm to obtain minimum cost spanning tree. Use Prim's algorithm to find the minimum cost spanning tree in the following graph:



8. Write and explain the Floyd Warshall algorithm to find the all pair shortest path. Use the Floyd Warshall algorithm to find shortest path among all the vertices in the given graph:

